

```
C:\Users\narva\OneDrive\Documentos\GITHUB\Tareas-Administracion de Base de Datos\Practica #4>docker compose up -d --build
#1 [internal] load local bake definitions
#1 reading from stdin 663B 0.0s done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 355B 0.0s done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/mysql:8.0
#3 DONE 0.2s

#4 [internal] load .dockerignore
#4 transferring context: 2B 0.0s done
#4 DONE 0.0s

#5 [internal] load build context
#5 transferring context: 101B done
#5 DONE 0.0s

#6 [1/4] FROM docker.io/library/mysql:8.0@sha256:7dcddc01f13bab2f15cde676d44d01f61fc9f99fe7785e86196dfc07d358ae2b
#6 resolve docker.io/library/mysql:8.0@sha256:7dcddc01f13bab2f15cde676d44d01f61fc9f99fe7785e86196dfc07d358ae2b
#6 resolve docker.io/library/mysql:8.0@sha256:7dcddc01f13bab2f15cde676d44d01f61fc9f99fe7785e86196dfc07d358ae2b 0.1s done
#6 DONE 0.1s

#7 [2/4] COPY 01_database.sql /docker-entrypoint-initdb.d/01_database.sql
#7 CACHED

#8 [3/4] COPY 02_tables.sql /docker-entrypoint-initdb.d/02_tables.sql
#8 CACHED

#9 [4/4] COPY 03_data.sql /docker-entrypoint-initdb.d/03_data.sql

#9 [4/4] COPY 03_data.sql /docker-entrypoint-initdb.d/03_data.sql
#9 CACHED

#10 exporting to image
#10 exporting layers 0.0s done
#10 exporting manifest sha256:5806e80f57b383394a12fff41815e1ad576cec75ffb2c6970e961c879329cd10 0.0s done
#10 exporting config sha256:fdde6b2ce0a7f42010f79b8c1d682b6f1c01524c90da0468bd7a14477927a7f3 0.0s done
#10 exporting attestation manifest sha256:a91198bee17964922cbb405c5696d26cd72adba869dc72a0ad53bbfe78ebacb8 0.0s done
#10 exporting manifest list sha256:0947d76bd8a9e94893591db86261ba70b73d4b604e712ceb6a0bc0bfc6aeffce
#10 exporting manifest list sha256:0947d76bd8a9e94893591db86261ba70b73d4b604e712ceb6a0bc0bfc6aeffce 0.0s done
#10 naming to docker.io/library/practica4-mysql_db:latest done
#10 unpacking to docker.io/library/practica4-mysql_db:latest 0.1s done
#10 DONE 0.4s

#11 resolving provenance for metadata file
#11 DONE 0.0s
[+] up 4/4
✓ Image practica4-mysql_db Built 2.7s
✓ Network practica4_default Created 0.1s
✓ Volume practica4_db_data_p4 Created 0.0s
✓ Container mysql_practica4 Started 0.6s

C:\Users\narva\OneDrive\Documentos\GITHUB\Tareas-Administracion de Base de Datos\Practica #4>docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
3bf9e6aa7223 practica4-mysql_db "docker-entrypoint.s..." 3 minutes ago Up 3 minutes 0.0.0.0:3309->3306/tcp, [::]:3309->3306/tcp mysql_practica4

C:\Users\narva\OneDrive\Documentos\GITHUB\Tareas-Administracion de Base de Datos\Practica #4>
```

MySQL

Driver for MySQL 8 and later

Ajustes de conexión

General

Metadatos

Errores y timeouts

Transferencia de datos

Data Editor

Editor SQL

General

Advanced

Driver properties

+ SSH, SSL, ...

No profile

Server

Connect by: ☒ Host ☐ URL

URL:

Server Host: Port:

Database: ☒ Show all databases

Authentication (Username/password)

Nombre de usuario:

Contraseña: ☒ Save password

Connection variables information

MySQL

Driver name: MySQL

Driver Settings

Licencia del driver

Probar conexión ...

Aceptar

Cerrar

Connection test

i

Conectado (52 ms)

Server: MySQL 8.0.46

Driver: MySQL Connector/J mysql-connector-j-8.2.0 (Revision: 06a1f724497fd81c6a659131fda822c9e5085b6c)

Aceptar

Detalles ↓

```
C:\Users\narva\OneDrive\Documentos\GITHUB\Tareas-Administracion de Base de Datos\Practica #4>docker exec -it mysql_practica4 mysql -uroot -psecret practica4_ecommerce
mysql: [Warning] Using a password on the command line interface can be insecure.
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
```

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 30
Server version: 8.0.46 MySQL Community Server - GPL
```

```
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```

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owners.
```

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
mysql> SHOW TABLES;
```

```
+-----+
| Tables_in_practica4_ecommerce |
+-----+
| compras                       |
| detalle_compra                |
| pagos                         |
| producto                     |
+-----+
4 rows in set (0.00 sec)
```

```
mysql> SHOW CREATE TABLE producto;
```

```
+-----+-----+
| Table      | Create Table
+-----+-----+
| producto  | CREATE TABLE `producto` (
  `id_producto` int NOT NULL AUTO_INCREMENT,
  `nombre` varchar(100) NOT NULL,
  `precio` decimal(10,2) NOT NULL,
  `stock` int NOT NULL,
  PRIMARY KEY (`id_producto`),
  CONSTRAINT `chk_producto_precio` CHECK ((`precio` > 0)),
  CONSTRAINT `chk_producto_stock` CHECK ((`stock` >= 0))
) ENGINE=InnoDB AUTO_INCREMENT=6 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> SHOW CREATE TABLE compras;
```

```
+-----+-----+
| Table | Create Table
+-----+-----+
| compras | CREATE TABLE `compras` (
  `id_compra` int NOT NULL AUTO_INCREMENT,
  `fecha` datetime NOT NULL DEFAULT CURRENT_TIMESTAMP,
  `estado` varchar(20) NOT NULL DEFAULT 'pendiente',
  `total` decimal(10,2) NOT NULL,
  PRIMARY KEY (`id_compra`),
  CONSTRAINT `chk_compras_estado` CHECK ((`estado` in (_latin1'pendiente',_latin1'completada',_latin1'cancelada'))),
  CONSTRAINT `chk_compras_total` CHECK ((`total` >= 0))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SHOW CREATE TABLE detalle_compra;
```

```
+-----+-----+
| Table | Create Table
+-----+-----+
| detalle_compra | CREATE TABLE `detalle_compra` (
  `id_detalle` int NOT NULL AUTO_INCREMENT,
  `id_compra` int NOT NULL,
  `id_producto` int NOT NULL,
  `cantidad` int NOT NULL,
+-----+-----+
```

```

-----+
| detalle_compra | CREATE TABLE `detalle_compra` (
  `id_detalle` int NOT NULL AUTO_INCREMENT,
  `id_compra` int NOT NULL,
  `id_producto` int NOT NULL,
  `cantidad` int NOT NULL,
  `precio_unitario` decimal(10,2) NOT NULL,
  PRIMARY KEY (`id_detalle`),
  KEY `fk_detalle_compra` (`id_compra`),
  KEY `fk_detalle_producto` (`id_producto`),
  CONSTRAINT `fk_detalle_compra` FOREIGN KEY (`id_compra`) REFERENCES `compras` (`id_compra`),
  CONSTRAINT `fk_detalle_producto` FOREIGN KEY (`id_producto`) REFERENCES `producto` (`id_producto`),
  CONSTRAINT `chk_detalle_cantidad` CHECK ((`cantidad` > 0)),
  CONSTRAINT `chk_detalle_precio` CHECK ((`precio_unitario` > 0))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+

```

1 row in set (0.00 sec)

```
mysql> SHOW CREATE TABLE pagos;
```

```

+-----+
| Table | Create Table
+-----+

```

```

+-----+
| pagos | CREATE TABLE `pagos` (
  `id_pago` int NOT NULL AUTO_INCREMENT,
  `id_compra` int NOT NULL,
  `monto` decimal(10,2) NOT NULL,
  `metodo` varchar(30) NOT NULL,
  `fecha` datetime NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id_pago`),
  KEY `fk_pagos_compra` (`id_compra`),
  CONSTRAINT `fk_pagos_compra` FOREIGN KEY (`id_compra`) REFERENCES `compras` (`id_compra`),
  CONSTRAINT `chk_pagos_monto` CHECK ((`monto` > 0))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+

```

1 row in set (0.00 sec)

```
mysql> SELECT * FROM producto;
```

```

+-----+
| id_producto | nombre                | precio  | stock |
+-----+
| 1 | Laptop Gamer          | 15999.90 | 10 |
| 2 | Mouse Inalambrico     | 299.50  | 50 |
| 3 | Teclado Mecanico      | 899.00  | 25 |
| 4 | Monitor 27 pulgadas  | 4599.00 | 5 |
| 5 | Audifonos Bluetooth   | 749.99  | 3 |
+-----+

```

5 rows in set (0.01 sec)

```

mysql> SELECT id_producto, nombre, stock FROM producto WHERE id_producto = 2;
+-----+-----+-----+
| id_producto | nombre           | stock |
+-----+-----+-----+
|          2 | Mouse Inalambrico |    50 |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO compras (estado, total) VALUES ('pendiente', 599.00);
Query OK, 1 row affected (0.01 sec)

mysql> SET @id_compra = LAST_INSERT_ID();
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE producto SET stock = stock - 2 WHERE id_producto = 2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> INSERT INTO detalle_compra (id_compra, id_producto, cantidad, precio_unitario) VALUES (@id_compra, 2, 2, 299.50);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO pagos (id_compra, monto, metodo) VALUES (@id_compra, 599.00, 'tarjeta');
Query OK, 1 row affected (0.00 sec)

mysql> UPDATE compras SET estado = 'completada' WHERE id_compra = @id_compra;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT id_producto, nombre, stock FROM producto WHERE id_producto = 2;
+-----+-----+-----+
| id_producto | nombre           | stock |
+-----+-----+-----+
|          2 | Mouse Inalambrico |    48 |
+-----+-----+-----+
1 row in set (0.00 sec)

```

```
mysql> SELECT * FROM compras;
```

id_compra	fecha	estado	total
1	2026-09-19 02:30:22	completada	599.00

```
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM detalle_compra;
```

id_detalle	id_compra	id_producto	cantidad	precio_unitario
1	1	2	2	299.50

```
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM pagos;
```

id_pago	id_compra	monto	metodo	fecha
1	1	599.00	tarjeta	2026-09-19 02:30:22

```
1 row in set (0.00 sec)
```

```
mysql> SELECT id_producto, nombre, stock FROM producto WHERE id_producto = 4;
```

id_producto	nombre	stock
4	Monitor 27 pulgadas	5

```
1 row in set (0.00 sec)
```

```
mysql> SELECT COUNT(*) AS compras_antes FROM compras;
```

compras_antes
1

```
1 row in set (0.01 sec)
```

```
mysql> START TRANSACTION;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> INSERT INTO compras (estado, total) VALUES ('pendiente', 45990.00);
```

```
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SET @id_compra = LAST_INSERT_ID();
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM compras WHERE id_compra = @id_compra;
```

id_compra	fecha	estado	total
2	2026-09-19 02:30:56	pendiente	45990.00

```
1 row in set (0.00 sec)
```

```
mysql> UPDATE producto SET stock = stock - 10 WHERE id_producto = 4;
```

```
ERROR 3819 (HY000): Check constraint 'chk_producto_stock' is violated.
```

```
mysql> SELECT id_producto, nombre, stock FROM producto WHERE id_producto = 4;
```

id_producto	nombre	stock
4	Monitor 27 pulgadas	5

```
1 row in set (0.00 sec)
```

```
mysql> SELECT COUNT(*) AS compras_despues FROM compras;
```

compras_despues
1

```
1 row in set (0.13 sec)
```

```

mysql> SELECT id_producto, nombre, stock FROM producto WHERE id_producto = 5;
+-----+-----+-----+
| id_producto | nombre           | stock |
+-----+-----+-----+
|          5 | Audifonos Bluetooth |     3 |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT COUNT(*) AS compras_antes FROM compras;
+-----+
| compras_antes |
+-----+
|             1 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT COUNT(*) AS pagos_antes FROM pagos;
+-----+
| pagos_antes |
+-----+
|             1 |
+-----+
1 row in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO compras (estado, total) VALUES ('pendiente', 749.99);
Query OK, 1 row affected (0.00 sec)

mysql> SET @id_compra = LAST_INSERT_ID();
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE producto SET stock = stock - 1 WHERE id_producto = 5;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> INSERT INTO detalle_compra (id_compra, id_producto, cantidad, precio_unitario) VALUES (@id_compra, 5, 1, 749.99);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO detalle_compra (id_compra, id_producto, cantidad, precio_unitario) VALUES (@id_compra, 5, 1, 749.99);
Query OK, 1 row affected (0.00 sec)

mysql> SELECT id_producto, nombre, stock FROM producto WHERE id_producto = 5;
+-----+-----+-----+
| id_producto | nombre           | stock |
+-----+-----+-----+
|          5 | Audifonos Bluetooth |     2 |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql> INSERT INTO pagos (id_compra, monto, metodo) VALUES (@id_compra, 0.00, 'tarjeta');
ERROR 3819 (HY000): Check constraint 'chk_pagos_monto' is violated.

```